Topology Over Parameters: How Circuit Connectivity Governs Learned Quantum Transform Bases for Image Compression

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2026-03-18

What matters more for a learned quantum transform basis—the number of parameters or the pattern of connections between qubits? We answer this question by relaxing the gate parameters of the Quantum Fourier Transform (QFT) tensor network and optimizing them on Riemannian manifolds ($U(2)$ for Hadamard gates, $U(1)^4$ for controlled-phase gates), preserving unitarity exactly at every iterate. Systematically comparing four circuit topologies—separable QFT, an Entangled QFT with cross-dimensional coupling, TEBD with ring connectivity, and a MERA-inspired hierarchical circuit—we uncover a striking result: *adding just n entanglement phases between row and column qubits outperforms doubling the parameter count via alternative topologies.* The Entangled QFT consistently achieves the best compression quality on both Quick Draw (32×32) and DIV2K (256×256) benchmarks, while TEBD—with comparable parameter count but local ring connectivity—underperforms even the fixed classical FFT by up to 1.5 dB. These findings establish that circuit connectivity topology, not parameter count or circuit depth, is the primary determinant of transform quality, with cross-dimensional entanglement capturing spatial correlations that no separable architecture can express. Our open-source Julia implementation provides a complete pipeline for learning and benchmarking parametric quantum circuit bases with Riemannian optimization and GPU acceleration.

Contributions

1. **Topology governs quality.** Through controlled experiments across four tensor network topologies, we demonstrate that circuit connectivity—not parameter count or circuit depth—is the dominant factor in learned transform quality. This finding has direct implications for quantum circuit architecture design.

2. **Entanglement breaks the separability barrier.** We introduce the Entangled QFT, which adds $O(\log N)$ cross-dimensional phase gates to the standard separable QFT. Despite this minimal overhead, it consistently outperforms all other topologies, including those with more parameters, by capturing spatial correlations invisible to separable transforms.

3. **End-to-end differentiable compression.** A frequency-weighted top-k truncation with a custom straight-through gradient estimator enables direct optimization of reconstruction quality through the non-differentiable sparsification step.

4. **Riemannian optimization on circuit manifolds.** We optimize circuit gates directly on $U(2) \times U(1)^4$ product manifolds via Cayley retraction and parallel-transported Adam momentum, guaranteeing exact unitarity without projection penalties.

5. **Open-source implementation.** The Julia library `ParametricDFT.jl` provides batched einsum contractions, Riemannian optimizers, and GPU acceleration, enabling reproducible research on parametric quantum transforms.

1 Introduction

The Discrete Fourier Transform has been the workhorse of signal compression for over half a century [1]. Yet despite its remarkable success, the DFT makes two assumptions that are manifestly wrong for natural images: it assumes periodic boundary conditions, and it treats spatial dimensions as independent. Every diagonal edge, every radial texture, every correlation between rows and columns is invisible to the separable $F_m \otimes F_n$ structure. Can we do better—not by abandoning the FFT's elegant tensor network structure, but by *learning* its parameters from data?

This paper begins from a simple observation: the Cooley–Tukey FFT is a tensor network of Hadamard gates H and controlled-phase gates $M_k = \mathrm{diag}(1, 1, 1, e^{i\pi/2^{k-1}})$. The classical DFT is one specific choice of gate parameters. By relaxing H to range over the full unitary group $U(2)$ and allowing the phases of M_k to become learnable, we obtain a continuous family of unitary transforms—a parameter manifold of which the DFT is a single point. The question becomes: *where on this manifold does the optimal transform for a given dataset lie, and what circuit structure gets us there?*

Answering this question leads to a surprising finding. We compare four circuit topologies of varying connectivity—separable QFT, Entangled QFT, TEBD, and MERA-inspired circuits—and discover that **topology trumps parameters**: the Entangled QFT, which adds just n cross-dimensional phase gates (8 real parameters for 256×256 images), consistently outperforms TEBD despite TEBD having comparable or greater parameter count. Meanwhile, TEBD's nearest-neighbor ring topology actually *degrades* performance below the fixed classical FFT, losing up to 1.5 dB in PSNR. The lesson is clear: in quantum circuit design for signal processing, *where* you place gates matters far more than *how many* you use.

Riemannian optimization on circuit manifolds. A naïve approach would optimize gate parameters in Euclidean space and periodically re-orthogonalize. This is both wasteful and numerically fragile. Instead, we exploit the Riemannian geometry of the parameter space: Hadamard gates live on $U(2)$, controlled-phase gates on $U(1)^4$, and the full circuit lives on their product manifold. Using Cayley retractions and parallel-transported Adam momentum [2], every iterate is *exactly* unitary—no projection penalties, no constraint violations, no accumulated numerical drift.

Related work. Tensor networks have been applied to classification [3], generative modeling [4], and data compression [5]. Parameterized quantum circuits underpin variational quantum algorithms [6, 7] and quantum machine learning [8, 9]. Riemannian optimization for unitary constraints appears in quantum control [10] and optimization on matrix manifolds [11]. Our work connects these threads in a new direction: we use quantum circuit topologies as *classical* tensor networks for image compression, discovering that the architecture of the circuit—not just its parameters—is the key design variable.

Outline. Section 2 reviews the tensor network representation of the FFT. Section 3 presents the four circuit topologies. Section 4 describes the loss functions, truncation scheme, and Riemannian optimizers. Section 5 reports benchmark results. Section 6 discusses implications and future directions.

2 Background: Tensor Network Representation of the FFT

We briefly review how the Cooley–Tukey FFT decomposes into a tensor network, establishing notation used throughout.

2.1 Cooley–Tukey decomposition

The DFT matrix of size $n = 2^k$ is $(F_n)_{ij} = \omega^{(i-1)(j-1)}$ where $\omega = e^{-2\pi i/n}$. The Cooley–Tukey algorithm decomposes this as

$$F_n \mathbf{x} = \begin{pmatrix} I_{n/2} & D_{n/2} \\ I_{n/2} & -D_{n/2} \end{pmatrix} \begin{pmatrix} F_{n/2} & 0 \\ 0 & F_{n/2} \end{pmatrix} \begin{pmatrix} \mathbf{x}_{\mathrm{odd}} \\ \mathbf{x}_{\mathrm{even}} \end{pmatrix}, \tag{1}$$

where $D_{n/2} = \mathrm{diag}(1, \omega, \omega^2, \ldots, \omega^{n/2-1})$ is the twiddle factor matrix.

2.2 Gate decomposition

In tensor network notation, a vector of size 2^k is a rank-k tensor with binary indices $q_0, q_1, \ldots, q_{k-1}$.

The butterfly structure of eq. (1) decomposes into:

- **Hadamard gates**: $H = \left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$, one per qubit.
- **Controlled-phase gates**: $M_k = \mathrm{diag}(1, 1, 1, e^{i\pi/2^{k-1}})$, connecting each pair of qubits at distance k in the recursive decomposition.

Recursive application yields k Hadamard gates and $k(k-1)/2$ controlled-phase gates for k qubits, evaluable in $O(2^k \cdot k)$ operations via optimized tensor contraction—matching the classical FFT complexity $O(N \log N)$ for $N = 2^k$.

For 2D images of size $2^m \times 2^n$, the separable 2D DFT is $F_m \otimes F_n$, requiring $m + n$ Hadamard gates and $m(m-1)/2 + n(n-1)/2$ controlled-phase gates applied independently to the row and column qubit registers.

3 Parametric Circuit Topologies

We now describe four circuit topologies, each defining a parametric unitary transform $\mathcal{T}(\boldsymbol{\theta}) : \mathbb{C}^{2^m \times 2^n} \to \mathbb{C}^{2^m \times 2^n}$. All topologies share the same gate parameterization: Hadamard-like gates $H \in U(2)$ and controlled-phase gates $M = \mathrm{diag}(1, 1, 1, e^{i\phi}) \in U(1)^4$ with learnable parameters. Setting all parameters to their QFT values recovers the classical DFT.

3.1 QFT basis

The QFT basis directly parameterizes the tensor network of section 2.2. Each gate becomes a learnable parameter: the $m + n$ Hadamard gates live on $U(2)$ and the $m(m-1)/2 + n(n-1)/2$ controlled-phase gates live on $U(1)^4$. The row and column circuits operate independently, yielding the separable transform $\mathcal{T} = F_m(\boldsymbol{\theta}_{\mathrm{row}}) \otimes F_n(\boldsymbol{\theta}_{\mathrm{col}})$.

Parameter count. For an image of size $2^m \times 2^n$: $(m+n)$ unitary gates on $U(2)$ (4 real parameters each) plus $m(m-1)/2+n(n-1)/2$ phase gates on $U(1)^4$ (1 real parameter each), giving $4(m+n) + m(m-1)/2 + n(n-1)/2$ total real parameters.

3.2 Entangled QFT basis

The separable QFT processes row and column dimensions independently, ignoring spatial correlations between them. Natural images exhibit strong cross-dimensional correlations (e.g., diagonal edges, radial textures). We propose an *Entangled QFT* that introduces controlled-phase gates coupling corresponding row and column qubits.

For a square image with n qubits per dimension, we add n entanglement gates $E_k = \mathrm{diag}(1, 1, 1, e^{i\phi_k})$ acting on qubit pairs (x_{n-k}, y_{n-k}) for $k = 1, \ldots, n$. Each gate multiplies a learnable phase $e^{i\phi_k}$ when both the row qubit x_{n-k} and column qubit y_{n-k} are in state $|1\rangle$.

The total transformation is:

$$\mathcal{T}_{\mathrm{entangled}} = U_{\mathrm{entangle}} \cdot (F_n \otimes F_n), \quad U_{\mathrm{entangle}} = \prod_{k=1}^{n} E_k. \tag{2}$$

This extension has several properties:

1. Computational complexity remains $O(N \log N)$ per dimension.
2. Only n additional real parameters are introduced ($O(\log N)$ overhead).
3. Setting $\phi_k = 0$ for all k recovers the separable QFT.

3.3 TEBD basis

Time-Evolving Block Decimation (TEBD) [12] uses a *ring topology* of nearest-neighbor controlled-phase gates preceded by Hadamard gates on each qubit. For n row qubits and n column qubits, each register forms an independent ring: qubit q_i is connected to q_{i+1} (with wrap-around $q_n \to q_1$).

Each gate is a controlled-phase gate $T_k = \mathrm{diag}(1, 1, 1, e^{i\phi_k})$ with a single learnable phase. The row ring has n gates and the column ring has n gates, giving $2n$ total learnable phases. The TEBD circuit evaluates in $O(n)$ gate operations per dimension, making it the computationally cheapest basis at inference time.

The parameter manifold is:

$$\mathcal{M}_{\mathrm{TEBD}} = \prod_{k=1}^{2n} U(1)^4. \tag{3}$$

3.4 MERA-inspired basis

The Multi-scale Entanglement Renormalization Ansatz (MERA) [13] uses hierarchical connectivity with *disentanglers* and *isometries* at increasing spatial scales. Our MERA-inspired basis borrows MERA's connectivity pattern while using controlled-phase gates (rather than full $U(4)$ unitaries) for consistency with the other bases and to preserve invertibility.

For $n = 2^k$ qubits, the circuit has $k = \log_2 n$ layers. Layer l has stride $s = 2^{l-1}$ and contains $n/(2s)$ pairs of disentangler and isometry gates. Both gate types use the same parameterization: $\mathrm{diag}(1, 1, 1, e^{i\phi})$ with one learnable phase per gate.

The total gate count per dimension is $2(n-1)$, and the circuit depth is $O(\log n)$. The hierarchical connectivity captures multi-scale features: layer 1 acts on nearest neighbors (fine scale), while deeper layers connect distant qubits (coarse scale).

3.5 Summary of circuit properties

Table 1: Comparison of the four circuit topologies for an image of size $2^n \times 2^n$ (n qubits per dimension).

Property	**QFT**	**Ent. QFT**	**TEBD**	**MERA**
Connectivity	all-to-all	all-to-all + XY	ring	hierarchical
Phase params	$n(n-1)$	$n(n-1)+n$	$2n$	$4(n-1)$
$U(2)$ gates	$2n$	$2n$	$2n$	$2n$
Circuit depth	$O(n^2)$	$O(n^2)$	$O(n)$	$O(n\log n)$
XY coupling	No	Yes	No	No

4 Training Objective and Optimization

4.1 Loss functions

We consider three loss functions with different trade-offs between sparsity promotion and reconstruction quality.

L_1 norm loss. $\mathcal{L}_{L_1}(\boldsymbol{\theta}) = \sum_{i,j} |\mathcal{T}(\boldsymbol{\theta})(\mathbf{x})_{i,j}|$. The ℓ_1 norm is the tightest convex relaxation of the ℓ_0 pseudo-norm, promoting sparse representations.

L_2 norm loss. $\mathcal{L}_{L_2}(\boldsymbol{\theta}) = \sum_{i,j} |\mathcal{T}(\boldsymbol{\theta})(\mathbf{x})_{i,j}|^2$. Encourages energy concentration with smoother gradients than L_1.

MSE reconstruction loss.

$$\mathcal{L}_{\mathrm{MSE}}(\boldsymbol{\theta}) = \|\mathbf{x} - \mathcal{T}(\boldsymbol{\theta})^{-1}(\mathrm{topk}(\mathcal{T}(\boldsymbol{\theta})(\mathbf{x}), k))\|_F^2, \quad (4)$$

where $\mathrm{topk}(\cdot, k)$ retains the k coefficients with the highest frequency-weighted scores and zeros out the rest. This loss directly optimizes reconstruction quality after truncation and is used in all experiments.

4.2 Frequency-weighted truncation

The top-k selection uses a frequency-weighted score rather than raw magnitude:

$$s_{i,j} = |y_{i,j}| \cdot (1 + w_{i,j}), \quad w_{i,j} = 1 - \frac{d_{i,j}}{2\, d_{\max}}, \quad (5)$$

where $d_{i,j} = \sqrt{(i-c_i)^2 + (j-c_j)^2}$ is the distance from frequency bin (i,j) to the DC component and $d_{\max}$ is the maximum such distance. Low-frequency coefficients receive up to a $2\times$ boost in their retention score, consistent with the human visual system's greater sensitivity to low-frequency content.

4.3 Straight-through gradient for top-k

The top-k truncation is non-differentiable: the set of retained indices changes discontinuously. We define a custom chain rule that passes gradients through retained coefficients and zeros out the rest—the *straight-through estimator* applied to truncation:

$$\frac{\partial\, \mathrm{topk}(\mathbf{y}, k)}{\partial y_{i,j}} = \begin{cases} 1 & \text{if } (i,j) \in \mathcal{S}, \\ 0 & \text{otherwise}, \end{cases} \quad (6)$$

where $\mathcal{S}$ is the set of retained indices. This is justified because (i) the true Jacobian has delta-function contributions at the boundaries that are unusable for gradient-based optimization, (ii) the mask is locally stable during training, and (iii) the gradient correctly conveys "adjust the retained coefficients to better approximate the signal."

4.4 Riemannian optimization

The circuit parameters live on two manifolds:

Definition 1 (Unitary manifold $U(2)$)**.** *The group of* 2×2 *unitary matrices with tangent space* $T_U U(2) = \{US : S^\dagger = -S\}$.

Definition 2 (Phase manifold $U(1)^4$)**.** *The product of four unit circles with tangent space* $T_{\mathbf{z}} U(1)^4 = \{i\boldsymbol{\theta} \circ \mathbf{z} : \boldsymbol{\theta} \in \mathbb{R}^4\}$.

Riemannian gradient. Given the Euclidean gradient $\nabla_E f$ from automatic differentiation:

$$\text{For } U(2): \quad \operatorname{grad} f(U) = U \cdot \operatorname{skew}(U^\dagger \nabla_E f), \tag{7}$$

$$\text{For } U(1)^4: \quad \operatorname{grad} f(\mathbf{z}) = i \cdot \operatorname{Im}(\bar{\mathbf{z}} \circ \nabla_E f) \circ \mathbf{z}, \tag{8}$$

where $\operatorname{skew}(A) = (A - A^\dagger)/2$ extracts the skew-Hermitian part.

Retraction. For $U(2)$, we use the Cayley retraction:

$$R_U(\alpha\xi) = \left(I - \frac{\alpha}{2}W\right)^{-1}\left(I + \frac{\alpha}{2}W\right)U, \tag{9}$$

where $W = \xi U^\dagger$ projected to $\mathfrak{u}(2)$ via $W \leftarrow (W - W^\dagger)/2$. The Cayley map preserves unitarity exactly for any step size α. For $U(1)^4$, we use normalization: $R_{\mathbf{z}}(\alpha\xi) = (\mathbf{z} + \alpha\xi)/|\mathbf{z} + \alpha\xi|$ elementwise.

Riemannian gradient descent with Armijo line search. The update rule is $x_{t+1} = R_{x_t}(-\eta \cdot \operatorname{grad} f(x_t))$, where η is determined by Armijo backtracking: starting from α_0, contract $\alpha \leftarrow \tau\alpha$ (with $\tau = 0.5$) until the sufficient decrease condition is satisfied:

$$\mathcal{L}(R_{x_t}(-\alpha \cdot \operatorname{grad} f(x_t))) \leq \mathcal{L}(x_t) - c \cdot \alpha \cdot \|\operatorname{grad} f(x_t)\|^2, \tag{10}$$

with $c = 10^{-4}$.

Riemannian Adam. We also implement Riemannian Adam [2], which maintains exponentially weighted first and second moment estimates of the Riemannian gradient with parallel transport of momentum between iterates. The bias-corrected update direction is $d_t = \hat{m}_t/(\sqrt{\hat{v}_t} + \epsilon)$, with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$. Momentum is transported between tangent spaces via projection:

$$\Gamma_{x_t \to x_{t+1}}(m_t) = \operatorname{proj}_{T_{x_{t+1}}\mathcal{M}}(m_t). \tag{11}$$

4.5 Batched einsum evaluation

The forward transform is implemented as an Einstein summation (einsum) contraction of the image tensor with the gate tensors. The contraction order is pre-optimized using a tree search algorithm (TreeSA) that minimizes the total operation count.

For batched training, a batch index β is appended to the image input/output indices:

$$Y_{q'_1,\ldots,q'_{m+n},\beta} = \sum_{q_1,\ldots,q_{m+n}} \left(\prod_j G^j_{\cdots}\right) X_{q_1,\ldots,q_{m+n},\beta}. \tag{12}$$

The gate tensors are shared across the batch, reducing GPU kernel launch overhead from $O(B)$ to $O(1)$.

5 Experiments

5.1 Setup

Datasets. We evaluate on two datasets of different scales:

- **Quick Draw** [14]: Hand-drawn sketches (28×28, zero-padded to 32×32, i.e., $m = n = 5$ qubits). Five categories (cat, dog, airplane, apple, bicycle), 1000+ images per category.
- **DIV2K** [15]: High-resolution natural photographs, preprocessed to 256×256 ($m = n = 8$ qubits). 800 training + 100 validation images.

Training configuration. All bases are trained using RiemannianGD with Armijo line search ($\alpha_0 = 0.01$, $c = 10^{-4}$, $\tau = 0.5$, max 10 backtracking steps). The loss function is $\mathcal{L}_{\text{MSE}}$ (eq. (4)) with $k = 10\%$ of total coefficients. We use a 20% validation split, early stopping with patience 2 epochs, and random seed 42 for reproducibility.

Metrics. We report PSNR (peak signal-to-noise ratio, $10\log_{10}(1/\text{MSE})$, higher is better) and SSIM (structural similarity index, higher is better) at compression ratios $k/N \in \{5\%, 10\%, 15\%, 20\%\}$. All metrics are averaged over the full evaluation set.

Baselines. We compare against the classical FFT (the untrained fixed-parameter DFT), which serves as the natural baseline since all learned bases reduce to it at their initialization point.

Table 2: Compression quality on Quick Draw (32×32, 5 qubits per dimension). PSNR (dB) / SSIM at four compression ratios. Best learned result in **bold**.

Method	**5%**	**10%**	**15%**	**20%**
FFT (fixed)	15.27 / 0.363	17.27 / 0.444	18.99 / 0.490	20.50 / 0.526
QFT (learned)	15.78 / 0.364	17.42 / 0.444	18.99 / 0.490	20.50 / 0.526
Ent. QFT (learned)	**15.38** / **0.379**	**17.42** / **0.458**	**19.15** / **0.505**	**20.70** / **0.541**
TEBD (learned)	14.57 / 0.340	16.35 / 0.414	17.77 / 0.465	19.11 / 0.501

5.2 Results on Quick Draw (32×32)

The results on Quick Draw (table 2) reveal a clear hierarchy driven by connectivity, not complexity. The Entangled QFT dominates across all compression ratios, achieving +0.20 dB over the classical FFT at 20% compression—a meaningful gain in the regime where every fraction of a dB matters for perceptual quality. More revealing is the *failure* of TEBD: despite having comparable parameter count to the Entangled QFT, its ring topology produces a −0.92 dB *deficit* relative to the fixed FFT at 10% compression. This is not merely underperformance—the optimizer actively finds a worse transform than the one it started from, because the ring topology cannot express the non-local frequency correlations that even simple sketch images exhibit. The separable QFT converges near the DFT parameters at higher ratios, confirming that the classical FFT is a strong local minimum on this manifold—but one that the Entangled QFT's cross-dimensional gates can escape.

5.3 Results on DIV2K (256×256)

Table 3: Compression quality on DIV2K (256×256, 8 qubits per dimension). PSNR (dB) / SSIM at four compression ratios. Best learned result in **bold**.

Method	**5%**	**10%**	**15%**	**20%**
FFT (fixed)	25.72 / 0.691	28.12 / 0.765	29.87 / 0.806	31.34 / 0.833
QFT (learned)	25.78 / 0.698	28.20 / 0.773	29.96 / 0.813	31.44 / 0.840
Ent. QFT (learned)	**25.79** / **0.699**	**28.20** / **0.773**	**29.96** / **0.813**	**31.45** / **0.840**
TEBD (learned)	24.49 / 0.614	26.61 / 0.685	28.25 / 0.729	29.70 / 0.765

Scaling to DIV2K (table 3) sharpens the picture. On high-resolution natural photographs, all QFT-family bases improve over the classical FFT, with the Entangled QFT maintaining its lead—though the margin narrows to +0.01 dB as the 8-qubit QFT structure already provides rich all-to-all connectivity within each dimension. The real story is TEBD's *catastrophic* scaling: its deficit grows from −0.92 dB on 32×32 sketches to −1.51 dB on 256×256 photographs. The ring topology's locality becomes increasingly punishing as image complexity grows, while the QFT's all-to-all structure and the Entangled QFT's cross-dimensional coupling gracefully handle the richer frequency content of natural scenes.

5.4 Training dynamics

Table 4: Training time (seconds) and convergence behavior on DIV2K.

Method	**Time (s)**	**Final train loss**	**Final val loss**
QFT	290.8	1162.0	2491.3
Ent. QFT	126.9	1162.0	2490.9
TEBD	277.8	1752.6	1441.2

An unexpected finding emerges from the training dynamics: the Entangled QFT converges *twice as fast* as the separable QFT (127 s vs. 291 s on DIV2K), despite having more parameters. This is not merely an implementation artifact—the entanglement gates create gradient pathways between the row and column circuits that do not exist in the separable architecture, effectively doubling the information flow per optimization step. TEBD tells a complementary story: it converges to a training loss 50% higher than the QFT variants (1752.6 vs. 1162.0), yet its validation loss is paradoxically *lower* than its training loss. The ring topology acts as an aggressive implicit regularizer—so aggressive that the model underfits, sacrificing reconstruction quality for a smoothness it was never asked to achieve.

5.5 Effect of circuit topology

The key experimental finding is that **circuit connectivity topology determines compression quality more than parameter count or circuit depth**:

1. The Entangled QFT adds only n phase parameters over the QFT (+8 parameters for DIV2K) yet consistently achieves the best or tied-best performance.

2. TEBD has $2n$ phase parameters—comparable to the Entangled QFT—but performs worst due to its local ring connectivity.
3. The QFT's all-to-all connectivity within each dimension closely matches the DFT's frequency structure, explaining why learned QFT bases converge near the DFT solution.
4. Cross-dimensional coupling (Entangled QFT) captures correlations that no amount of intra-dimensional parameters can express in a separable architecture.

6 Discussion

The FFT as a strong local minimum. A natural question is why the learned bases do not dramatically exceed the classical FFT. Our results provide a geometric answer: the DFT occupies a *deep local minimum* on the parameter manifold. The separable QFT's learned parameters converge close to DFT values, confirming that Cooley and Tukey's 1965 design [1] is near-optimal within the separable family for generic images. The Entangled QFT improves upon this not by finding a better point in the same basin, but by *expanding the search space* with cross-dimensional connections that the separable manifold cannot reach—analogous to how wavelets improve on the DFT not by better frequency resolution, but by introducing spatial localization.

Cross-dimensional entanglement as a design principle. The Entangled QFT's success has implications beyond this specific application. In quantum circuit design, the conventional wisdom is that more parameters or deeper circuits yield better expressiveness. Our results challenge this: the Entangled QFT's n cross-dimensional phases (8 parameters for 256×256 images) outperform TEBD's $2n$ ring-topology phases because they connect the *right* degrees of freedom. The entanglement gates allow the basis to represent frequency components that mix row and column information—analogous to diagonal Gabor-like features that no separable architecture can express, regardless of parameter count. This suggests a broader principle for variational quantum circuits: *topology-aware architecture design guided by the symmetry structure of the target problem is more valuable than blindly increasing circuit depth or width.*

Implications for quantum computing. While our circuits are evaluated classically as tensor networks, the parametric QFT structure maps directly to quantum hardware. A quantum computer implementing the Entangled QFT would require only n additional CPHASE gates beyond the standard QFT circuit—a negligible overhead on current hardware where two-qubit gate fidelity is the primary bottleneck. As quantum processors scale, our framework provides a principled way to design application-specific transform circuits optimized for the data they will process.

Limitations and future directions. Our benchmarks currently compare against the classical FFT; comparisons with the DCT, wavelets, and learned dictionaries (K-SVD) would locate parametric quantum bases within the broader compression landscape. We anticipate that the advantages may be more pronounced on structured image classes (e.g., self-similar textures, medical imaging) where domain-specific correlations can be exploited by the learned gates. Ablation studies isolating the contribution of Riemannian vs. projected Euclidean optimization, and evaluation of the MERA-inspired basis on power-of-two image patches, are natural next steps.

Reproducibility. All experiments are implemented in the open-source Julia library `ParametricDFT.jl`, which provides the full training pipeline including batched einsum evaluation, Riemannian optimizers, checkpointing, and optional GPU acceleration via CUDA. Trained circuit parameters, benchmark scripts, and evaluation code are included in the repository.

7 Conclusion

We set out to ask whether relaxing the parameters of the FFT's tensor network could yield better compression bases, and discovered something more fundamental: *the topology of the circuit matters more than the number of its parameters.* The Entangled QFT—which adds just $O(\log N)$

cross-dimensional gates to the classical QFT—consistently outperforms topologies with comparable or greater parameter counts, while TEBD's local ring connectivity actively degrades below the fixed FFT baseline.

This finding carries a practical message for both classical signal processing and quantum circuit design. For signal processing: separable transforms leave cross-dimensional correlations on the table, and even minimal entanglement between dimensions can recover them. For quantum circuits: architecture search should prioritize connectivity patterns matched to problem structure over raw circuit depth. Riemannian optimization on the natural $U(2) \times U(1)^4$ parameter manifold makes this search both principled and practical, preserving unitarity exactly while enabling gradient-based exploration of the transform landscape.

The parametric quantum transform framework introduced here is deliberately extensible. New circuit topologies can be defined by specifying a connectivity pattern; the manifold optimization, batched einsum evaluation, and differentiable truncation machinery apply unchanged. We believe this opens a productive direction at the intersection of tensor networks, Riemannian geometry, and data-driven signal processing.

Acknowledgments

References

[1] James W Cooley and John W Tukey. An algorithm for the machine calculation of complex Fourier series. *Mathematics of Computation*, 19(90):297–301, 1965.

[2] Gary Bécigneul and Octavian-Eugen Ganea. Riemannian adaptive optimization methods. In *International Conference on Learning Representations*, 2019.

[3] Edwin Stoudenmire and David J Schwab. Supervised learning with tensor networks. *Advances in Neural Information Processing Systems*, 29, 2016.

[4] Zhao-Yu Han, Jun Wang, Heng Fan, Lei Wang, and Pan Zhang. Unsupervised generative modeling using matrix product states. *Physical Review X*, 8(3):031012, 2018.

[5] Ivan V Oseledets. Tensor-train decomposition. *SIAM Journal on Scientific Computing*, 33(5):2295–2317, 2011.

[6] Marco Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J Coles. Variational quantum algorithms. *Nature Reviews Physics*, 3(9):625–644, 2021.

[7] Jarrod R McClean, Jonathan Romero, Ryan Babbush, and Alán Aspuru-Guzik. The theory of variational hybrid quantum-classical algorithms. *New Journal of Physics*, 18(2):023023, 2016.

[8] Maria Schuld and Nathan Killoran. Quantum machine learning in feature Hilbert spaces. *Physical Review Letters*, 122(4):040504, 2019.

[9] Marcello Benedetti, Erika Lloyd, Stefan Sack, and Mattia Fiorentini. Parameterized quantum circuits as machine learning models. *Quantum Science and Technology*, 4(4):043001, 2019.

[10] Navin Khaneja, Timo Reiss, Thomas Schulte-Herbrüggen, and Steffen J Glaser. Optimal control of coupled spin dynamics: design of NMR pulse sequences by gradient ascent algorithms. *Journal of Magnetic Resonance*, 172(2):296–305, 2005.

[11] P-A Absil, Robert Mahony, and Rodolphe Sepulchre. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2008.

[12] Guifré Vidal. Efficient simulation of one-dimensional quantum many-body systems. *Physical Review Letters*, 93(4):040502, 2004.

[13] Guifré Vidal. Entanglement renormalization. *Physical Review Letters*, 99(22):220405, 2007.

[14] Google. Quick, draw! dataset. https://quickdraw.withgoogle.com/data, 2017.

[15] Eirikur Agustsson and Radu Timofte. NTIRE 2017 challenge on single image super-resolution: Dataset and study. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops*, pages 126–135, 2017.

A Riemannian optimization details

A.1 Cayley retraction derivation

For a unitary matrix $U \in U(2)$ and tangent vector $\xi = US$ where $S^\dagger = -S$, the Cayley retraction maps:

$$W = \xi U^\dagger = USU^\dagger, \tag{13}$$

$$W \leftarrow \frac{W - W^\dagger}{2} \quad (\text{project to } \mathfrak{u}(2)), \tag{14}$$

$$R_U(\alpha\xi) = \left(I - \frac{\alpha}{2}W\right)^{-1}\left(I + \frac{\alpha}{2}W\right)U. \tag{15}$$

Since W is skew-Hermitian, $(I - \frac{\alpha}{2}W)^{-1}(I + \frac{\alpha}{2}W)$ is unitary for any $\alpha \in \mathbb{R}$, as can be verified by checking that the product with its adjoint equals the identity.

For 2×2 matrices, the inverse admits a closed form via the adjugate:

$$(I - \frac{\alpha}{2}W)^{-1} = \frac{1}{\det(I - \frac{\alpha}{2}W)} \operatorname{adj}(I - \frac{\alpha}{2}W). \tag{16}$$

A.2 Parallel transport for Riemannian Adam

For $U(2)$, the projection-based transport from U_t to U_{t+1} is:

$$\Gamma_{U_t \to U_{t+1}}(m_t) = U_{t+1} \cdot \operatorname{skew}(U_{t+1}^\dagger m_t). \tag{17}$$

For $U(1)^4$, the transport is the tangent projection at the new point:

$$\Gamma_{\mathbf{z}_t \to \mathbf{z}_{t+1}}(m_t) = i \cdot \operatorname{Im}(\bar{\mathbf{z}}_{t+1} \circ m_t) \circ \mathbf{z}_{t+1}. \tag{18}$$

Both are exact to first order in the step size.

B Detailed benchmark configurations

Table 5: Hyperparameters used in the benchmark experiments.

Parameter	**Value**
Optimizer	RiemannianGD
Initial learning rate α_0	0.01
Armijo constant c	10^{-4}
Backtracking factor τ	0.5
Max backtracking steps	10
Loss function	MSE ($k = 10\%$)
Validation split	20%
Early stopping patience	2 epochs
Random seed	42
Epochs (Quick Draw)	5
Training images (Quick Draw)	20
Epochs (DIV2K)	5
Training images (DIV2K)	20